

Color-Corrected Full Ray-Based Computer-Generated Holography

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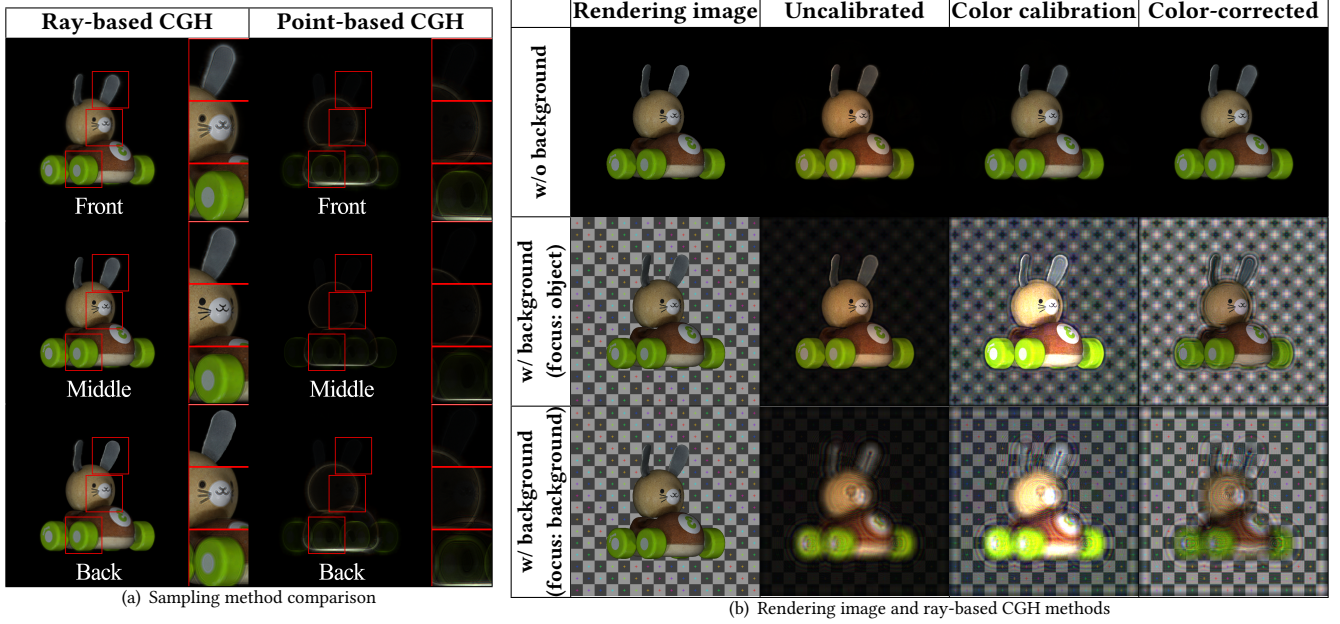


Figure 1: (a) Comparison between Ray-based CGH and Point-based CGH, which display numerical reconstruction results focused on the front, middle, and back. The number of points in point-based CGH is 4×10^6 (b) Comparison of rendering image and holograms uncalibrated (Eq. 1), color calibration (Eq. 2 w/o amplitude factors $z_p(1 - \rho^3)$), and color-corrected method (Eq. 2). The object and background are placed at -1.44 cm and -4 cm, respectively. The hologram resolution is 1024×1024 , the pixel pitch is $3.6 \mu\text{m}$, and wavelengths are 450 nm, 532 nm, and 638 nm.

Abstract

Ray-based computer-generated holography can generate holograms with precise occlusion. However, its naive point-to-point wave propagation introduces noticeable visual artifacts, such as color imbalance and extreme brightness variations, due to suboptimal ray sampling that disregards differences between wavelengths. To address this problem, we propose color-corrected ray-based CGH that calibrates color using reference planes and adjusts amplitude

with correction factors. The results show that our color-corrected ray-based CGH achieves a consistent color to a rendered image.

CCS Concepts

• **Computing methodologies** → *Rendering; Ray tracing*; • **Hardware** → Emerging optical and photonic technologies.

Keywords

Computer-generated holography, Hologram, Ray tracing, Color-calibration

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1 Introduction

Ray-based CGH samples point light sources (PLSs) using ray-tracing and propagates the light waves from the sampled PLSs to hologram [Ichikawa and Sakamoto 2012]. It usually employs point-to-point wave propagation [Maimone et al. 2017] that each pixel of a hologram is determined by superposition of propagated wave from PLSs. Unlike point-based CGH, which propagates static PLSs acquired from a single view, the ray-based CGH propagates unique PLSs generated by ray-tracing for each hologram pixel enables accurate occlusion handling without explicit wave masking methods [Matsushima and Nakahara 2009].

However, since uniform ray-sampling of ray-based CGH occurs visual artifacts such as color-imbalance, exaggerated brightness and high-frequency error because of suboptimal PLS sampling, even though it achieves accurate occlusion handling. To fix these visual artifacts, we propose color-corrected ray-based CGH which calibrates color channel by reference plane generated by max dynamic range and adjusts amplitude by distance. As a result, our method can generate holograms which have the same color distribution as a rendering target.

2 Our approach

Our color-corrected method is based on point-to-point CGH given by:

$$H_c(x, y) = \sum_{p \in P_{x,y}} \frac{A_p}{r} \exp(-\sqrt{-1}k_c(r - z_p)) \quad (1)$$

$$r = \sqrt{(x - x_p)^2 + (y - y_p)^2 + (z_p)^2}$$

$H_c(x, y)$ denotes the hologram plane of color channel c ; p is a point light source in point set $P_{x,y}$; A_p is amplitude of p ; r is the physical distance between a hologram pixel located at $(x, y, 0)$ and p located at (x_p, y_p, z_p) ; and k_c is wavenumber of color channel c . The point-based CGH has the same P for every pixel but the ray-based CGH has a unique $P_{x,y}$ for each pixel of the hologram, (x, y) .

The problem is because Eq. 1 with $P_{x,y}$ acquired by uniform ray sampling where ray directions are evenly distributed within the maximum diffraction angle $[-\theta_{max}/2, \theta_{max}/2]$ causes color-imbalance and exaggerated brightness. This is because 1 applies same dynamic range and ray distribution for each wavelength to reduce computation cost, even though different wavelength has different diffraction angles and phase variations.

To address this problem, we propose color-corrected ray-based CGH given by Eq. 2 that calibrates a hologram by reference plane.

$$H_c(x, y) = \frac{\sum_{p \in P_{x,y}} A_p z_p (1 - \rho^3) \exp(-\sqrt{-1}k_c(r - z_p))}{\text{avg.}(|R_c|_{amp.})} \quad (2)$$

$$R_c(x, y) = \sum_{p \in P_{x,y}} 1 \cdot \exp(-\sqrt{-1}k_c(r - z_p)) \quad (3)$$

$$\rho = \begin{cases} \frac{\sqrt{(\theta_x)^2 + (\theta_y)^2}}{\theta_{max}}, & (\theta_x)^2 + (\theta_y)^2 < (\theta_{max})^2 \\ 1, & \text{otherwise.} \end{cases} \quad (4)$$

The color reference plane, $R_c(x, y)$, is generated by PLSs with same amplitudes but it has the same color difference as $H_c(x, y)$ because it has the phase values of PLSs are same. Thus, dividing

$H_c(x, y)$ by average of $R_c(x, y)$ calibrate the colors, correcting imbalances between channels.

Although the colors of a hologram are calibrated, the brightness can still appear excessively bright or dark when the scene depth range is large. To address this issue, we amplify the amplitude based on the distance between PLS and hologram plane unlike the attenuation used in Eq. 1. Also, we found that the hologram contains high-frequency noise even though its ray distribution is limited by the diffraction angle. In order to reduce the noise, we apply a low-pass filter with ρ which is defined based on ray direction. θ_x and θ_y denotes x- and y-angle of the rays, which are uniformly distributed in $[-\theta_{max}/2, \theta_{max}/2]$. In our heuristic studies, a cubic function produces the best visual quality in our method.

3 Results

We implemented ray-based CGH methods: Eq. 1, Eq. 2, and it without amplitude handling(color calibration only). The implementation employs OptiX 9.0, where hologram plane and reference plane are computed within a single OptiX shader on a Geforce RTX 5080 GPU. calculating average of reference plane is performed on Intel i7-12700K CPU within multi-threaded. To enable the best hologram quality, we set the sample-per-pixel to 1024^2 . The Eq. 1 and Eq. 2 takes 45 s and 47 s, respectively, for a single object, and 62 s and 63 s, respectively, for an object with a background.

The reconstruction results are shown in Fig. 1. The uncalibrated method produces an imbalanced color distribution, with the red channel values higher than the others, and its background appears too dark. The color calibration method using R_c achieves a color distribution similar to the rendered image, but the object appears overly bright when the hologram is generated with a background. Our color corrected method that multiplies the amplitude by $z_p(1 - \rho^3)$ achieves both the color distribution and the brightness consistent with the rendered image.

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